R98: finite denominator support of the degree-16 marked-23 lambda equation

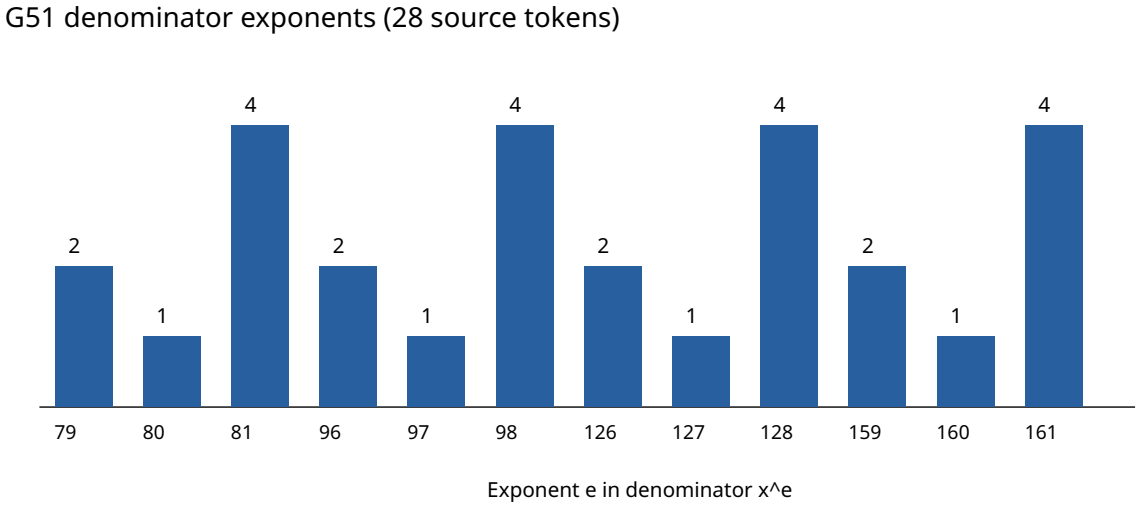
Koblitz modular IC research

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The source-bound audit of the accepted R51/R63 degree-112 triangular donor found 28 denominator tokens in the actual G_{51} equation. Every token is x^e, and the maximum exponent is 161. Thus $x^{161}G_{51}$ has polynomial coefficients in $\mathbf{F}_2[x, y, l]$; the displayed monogenic equation has no finite coefficient poles away from $x = 0$. The source file has 135,924 bytes and SHA-256 `7a92cf0f1665b80bef5a38c57ca45136e530bfcf8cd13a430942af75684b7409`.

The checker accounted for every division slash inside the unique G_{51} assignment. A planted $(x + 1)$ denominator was rejected; a separate Python process reproduced the JSON byte for byte. The Lean 4.34.1 check proves that all 28 recorded exponents are at most 161, that 161 occurs, and that each has a complementary exponent summing to 161. This Lean proof checks the reported list; the source-binding and token extraction are in the Python checker.

Attempt 01 failed its negative control because the mutation landed in an earlier assignment in the one-line donor. The failed attempt is recorded and amendment 02 confines the mutation to G_{51}.



The next calculation is the norm of the lambda derivative resultant, $\mathrm{Res}_y(f_{51}, \mathrm{Res}_l(G_{51}, \partial_l G_{51}))$, on $D(x)$. Its zero support, together with the R97 degree-7 base-order discriminant support and the fiber at infinity, will identify fibers for local normalization. This report makes no normalized ramification claim.